Krishna Mehra

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Education

NIT Surat	B.Tech - Artificial Intelligence	CGPA: 7.4	2023–Present
SGVP, Rajkot	12th (CBSE)	87.1%	2023
Kendriya Vidyalaya, Rajkot	10th (CBSE)	91.2%	2021

Research

- **Can You Break RLVER? Probing Adversarial Robustness of RL-Trained Empathetic Agents**
  Reinforcement Learning, LLM Evaluation, Empathy Modeling
  - Designed adversarial personas and psychologically grounded dialogue trajectories for the **Adversarial Empathy Benchmark (AEB)**, including gaslighting, emotional escalation, and manipulation-based interactions.
  - Conducted controlled evaluations across **480 adversarial dialogues** comparing RL-trained (PPO, GRPO) and baseline LLMs under identical conditions.
  - Proposed the **Emotional Consistency Score (ECS)** to distinguish emotional state tracking from state improvement in empathetic dialogue systems.
  - Observed improved emotional responsiveness, stronger hidden intention detection, and zero dialogue collapse in RL-trained agents under adversarial settings.
  - **Manuscript under review at NeurIPS 2026.**

Projects

- **VidSynth AI – YouTube Chat Assistant** *GitHub* | *Live Demo*
  - Independently developed the complete **VidSynth AI** system, including transcript processing, RAG pipeline integration, chatbot workflow, and deployment pipeline.
  - Developed a **Streamlit-based RAG application** using **Gemini 2.5 Flash Lite** and **ChromaDB** for context-aware YouTube video question answering.
  - Engineered transcript extraction, English translation, topic detection, and note generation modules using **LangChain** and **Sentence Transformers**.
  - Designed and deployed a real-time interactive web application on **Hugging Face** for video summarization and conversational querying.
- **Rare Gen Master - Synthetic Data Generation for Rare Cancer Diseases Using GANs**
  Deep Learning, PyTorch GitHub
  - Implemented a **GAN-VAE framework** on the Brain Glioma dataset to synthesize rare cancer imaging data, addressing dataset scarcity and class imbalance.
  - Validated generated samples using **FID/IS metrics** and evaluated downstream classification performance improvements.
  - Focused on generating privacy-preserving synthetic medical data for safe healthcare ML experimentation.
  - **Research paper accepted at IEEE AICAS 2026** for this work on GAN-VAE based synthetic rare cancer data generation for rare brain cancer imaging datasets.

Skills

Programming Languages: C, C++, Python
Web Development: HTML
Databases: MongoDB, MySQL
AI Domains: Machine Learning, Deep Learning, Reinforcement Learning, NLP, RAG Systems, GANs, VAEs
AI/ML Libraries & Frameworks: PyTorch, Scikit-learn, TensorFlow, Hugging Face Transformers, LangChain, ChromaDB
Data Science & Visualization: Pandas, NumPy, Matplotlib, Seaborn
Tools & Platforms: Git, Colab, Kaggle, GitHub, Jupyter Notebook, Anaconda, VS Code

Positions of Responsibility

- **Manager**, *MINDBEND (Annual Technical Festival)* **Jan 2026 – Apr 2026**
- **Co-Head**, *MINDBEND (Annual Technical Festival)* **Jan 2025 – Apr 2025**
- **Executive**, *MINDBEND (Annual Technical Festival)* **Jan 2024 – Apr 2024**